USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 5 LAB - First Flowgraphs
Lab Packet
Spring 2027

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 2.3: I can install GNU Radio and use GNU Radio Companion (GRC) to build, inspect, and run a basic flowgraph.
Learning Objective 2.4: I can capture a real RF signal using an SDR and interpret its time- and frequency-domain representations.

This packet is the L5 Lab, in three parts, and you turn it in once, complete, before Lesson 6. Do **Part 1** on your own before the lab period: install GNU Radio and work out the numbers your block settings fix. **Part 2** is your own measurements and observations, with your own laptop and RTL-SDR, recorded in class. Do **Part 3** on your own after the lab: explain what you saw.

Grading. One submission, graded against Objectives 2.3 and 2.4 on the course's 1 / 0 scale, with revision allowed under course policy. Paste screenshots where a question asks for one.

Receive only. Nothing in this lab transmits. The RTL-SDR cannot transmit, and the tone flowgraph never leaves your laptop.

The flowgraphs and the Python script are on the L5 lesson page: `L05-tone.grc`, `L05-fm-spectrum.grc`, and `L05-psd.py`.

Part 1: Pre-Lab (individual, before the lab period)

1. **Install and verify.** Install radioconda as the L5 lesson page describes, plug in your RTL-SDR, and run the commands below from the radioconda prompt.

   (a) Run `gnuradio-config-info --version` and record the version.

   version = _____

   (b) Run `rtl_test` and record the tuner it reports. Stop it with Ctrl-C after a few seconds.

   tuner = _____

   (c) Open `L05-tone.grc` in GRC, run it, and paste a screenshot of the running flowgraph's window.

2. Numbers from your settings. Quick arithmetic from the block settings; you will check each on the screen at the bench.

(a) The tone flowgraph samples at $f_s = 32$ kHz. Work out where the frequency sink shows each tone.

Tone (kHz)	1	12	20	30	-25
Expected (kHz)					

(b) The FM flowgraph samples at 2.4 MS/s centered on 100.1 MHz. Find the band the display shows.

band =

(c) Find the size of a 2 s recording at 2.4 MS/s.

Part 2: Measurements (your own laptop and RTL-SDR, in class)

3. **Build the tone flowgraph.** Start GRC with an empty canvas. Do not open `L05-tone.grc`: build it yourself from the Options block, a `samp_rate` variable of 32000, a QT GUI Range named `freq` from -48000 to 48000, a complex Signal Source, a Throttle, a QT GUI Time Sink, and a QT GUI Frequency Sink.

   (a) Paste a screenshot of your flowgraph on the GRC canvas. Every connection must be black, not red.

   (b) Set each tone with the slider and record where the spike appears.

Tone (kHz)	1	12	20	30	-25
Measured (kHz)					

4. The FM band. Connect the antenna to the RTL-SDR. Find a station in GQRX first, then look at the band in GNU Radio.

(a) Run `gqrx`, choose the RTL-SDR, set the mode to WFM (stereo), and press the power button. Click across the FM band until a station plays clearly, and record its frequency. Then quit GQRX. GQRX comes with radioconda; if GQRX does not start on your laptop, SDR++ has the same controls.

station = _____

(b) Run `L05-fm-spectrum.grc` with its center frequency set to the station from part (a). Then tune across 88 to 108 MHz and record the three strongest stations: the frequency, the peak level on the display, and the approximate width.

	Frequency (MHz)	Peak (dB)	Width (kHz)
Station 1			
Station 2			
Station 3			

(c) Find the DC spike. Tune so that a station sits on it, then tune 300 kHz away. Describe what moves and what does not.

(d) With the strongest station on screen, set the gain to each value and record the noise floor away from any station. Mark the first setting where extra signals appear between stations.

Gain (dB)	0	20	30	40	49
Floor (dB)					
Spurs?					

5. **Record and plot.** Center the strongest station slightly off the DC spike and set the highest gain in your 4(d) table at which no spurs appeared.

   (a) Enable the File Sink, run the flowgraph for about 2 s, stop it, and record the file size.

   size = _____

   (b) Run the plotting script with your center frequency, and paste the plot:

   ```
   python L05-psd.py fm_capture.cfile --fs 2.4e6 --fc <center Hz>
   ```

Part 3: Writeup (individual, after the lab)

6. Analysis.

(a) Compare your measured aliases in Question 3(b) with your answers in Question 2(a). Explain why a 20 kHz tone and a -12 kHz tone produce the same samples at 32 kHz.

(b) Using your table in Question 4(d), describe in words what happened as you raised the gain from 0 to 49 dB: to the noise floor, to how far the strongest station stood above it, and to the space between stations.

(c) State the gain you would use to receive weak stations in this room, and justify it from your table in Question 4(d).

Documentation: _____